Anjani Sarawagi

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.39
Matrix High School, Sikar	Class XII (RBSE)	2023	88.00%
Adarsh Vidya Mandir School, Taranagar	Class X (RBSE)	2021	95.00%

EXPERIENCE

Research Intern | Megha Lab, IIT Hyderabad

Satellite Based Rainfall Nowcasting Using Spatiotemporal U Net: Impact of Loss Function Selection | IEEE InGARSS 2026

- Built a Spatiotemporal U Net with a convolutional encoder decoder and ConvLSTM bottleneck, trained on 25,000 satellite rainfall sequences from 2000 to 2025 to forecast rainfall two hours ahead over the Central India Monsoon Zone
- Designed and benchmarked three loss functions to counter class imbalance in rainfall intensity, where weighted MSE cut test RMSE by 12.7 percent over standard MSE while a focal loss variant more than doubled heavy rainfall detection skill

Sharpening Precipitation Nowcasts with Deep Learning: Architecture Dependent Effects of Loss Functions | NeurIPS 2026 Workshop

- Investigated whether frequency domain supervision improves forecast sharpness, training ConvLSTM and Spatiotemporal U Net on NASA IMERG satellite precipitation under both Fourier and pixel wise losses across five evaluation metrics
- Found the benefit to be architecture dependent, raising Spatiotemporal U Net structural similarity from 0.13 to 0.65 and fraction skill score from 0.006 to 0.667 while ConvLSTM saw negligible change, showing loss and architecture must be co designed

SKILLS

Programming Languages: Python, C, C++, SQL, JavaScript, HTML, CSS

Machine Learning: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, Matplotlib

Web and Backend: Flask, Streamlit, React, Node.js, Express.js, REST APIs

Databases: MongoDB, PostgreSQL, SQLite, Supabase

Core CS: Data Structures and Algorithms, Object Oriented Programming, DBMS, System Design

PROJECTS

Neural Style Transfer | Real Time AI Art Generation Web App

Python, PyTorch, Flask, VGG19, Computer Vision, Deep Learning, NumPy, Pillow, Gunicorn

- Implemented the AdaIN research paper from scratch in PyTorch to build a real time neural style transfer model, training a custom decoder network over 200 epochs on the COCO2017 dataset with a frozen VGG19 encoder for feature extraction
- Designed a composite loss function combining content reconstruction loss with multi layer style loss computed across four VGG feature maps, achieving arbitrary style transfer in a single forward pass instead of costly per image optimization

- Deployed the trained model as a Flask web application with image upload, adjustable style strength blending, and live result preview, configured for production serving through Gunicorn

AI Gym Coach | Real Time Pose Estimation and LLM Voice Coaching

Python, MediaPipe, OpenCV, Streamlit, WebRTC, Groq API, Llama 3.3, SQLite, Computer Vision, Generative AI

- Built a real time fitness coaching application using MediaPipe pose landmark detection to track 33 body keypoints from live webcam feed, implementing joint angle based rep counting and form validation across five different exercises
- Integrated a Groq hosted Llama 3.3 model with custom prompt engineering to generate contextual coaching cues, converted to speech through gTTS for hands free audio feedback during workouts
- Designed the exercise detection layer using an abstract base class with per exercise strategy implementations, and handled concurrent video processing with thread safe state synchronization between the WebRTC stream and the Streamlit interface

SnapClass | AI-Based Classroom Attendance System

Python, Streamlit, dlib, scikit-learn, Resemblyzer, Librosa, Supabase, PostgreSQL, Git

- Engineered a full-stack computer vision and speaker recognition system using dlib face embeddings and SVM classification to identify multiple students simultaneously from a single classroom photo or audio recording
- Designed and deployed a normalized PostgreSQL database on Supabase for teachers, students, subjects, and attendance records, implementing bcrypt password hashing and QR code based student self-enrollment
- Developed role-based teacher and student dashboards in Streamlit with subject management, attendance tracking, and per-subject analytics, and configured the complete Python environment and deployment workflow

Zerodha Clone | Full Stack Stock Trading Platform

MongoDB, Express.js, React, Node.js, Mongoose, Axios, Chart.js, REST API, MERN Stack

- Architected a full stack MERN trading platform replicating Zerodha, delivering an interactive dashboard for portfolio holdings, positions, and order execution, plus a responsive multi page marketing website
- Engineered RESTful APIs in Express.js with Mongoose schemas to model and persist holdings, positions, and buy or sell orders in MongoDB, consumed asynchronously by the React dashboard through Axios for seamless client server communication
- Integrated interactive Chart.js visualizations for portfolio distribution and stock performance analytics, and implemented React Context API for centralized state management driving the order placement workflow across components

RELEVANT COURSES

Computer Science: Data Structures and Algorithms, Introduction to Programming, Machine Learning, Deep Learning

Mathematics: Linear Algebra, Probability and Statistics, Calculus, Differential Equations, Transform Techniques, Complex Numbers

POSITIONS OF RESPONSIBILITY

Reviewer, IEEE InGARSS 2026 - Served on the technical review committee for an IEEE geoscience conference

Core Team, ML Domain, Tinkerers Lab, IITH - Led Machine Learning projects and mentored juniors

Teaching Assistant, Introduction to Programming, IITH - Conducted lab sessions and designed assignments

Core Team, Vibes Music Club, IITH - Organized campus music events and performed at cultural fests

Volunteer, National Service Scheme, IITH - Taught in rural schools and led campus awareness drives